HYPERTENSION MONITORING PROTOTYPE

PULSE: Data & AI

What we built, what it shows, and what is still missing

Data + AI/ML workstream · 30 October 2026

Prototype: all patient data is simulated and labelled. Real public datasets are used only to calibrate and check.

THE IDEA

Data flows to the doctor; the AI helps them see earlier

1 · Measurements

- Blood-pressure cuff readings
- Ring / bracelet: heart rate, HRV, SpO2, skin temperature
- Symptoms, lifestyle, medication doses

→

2 · Secure database

- Supabase (Postgres 17) in the cloud
- Every row checked on the way in
- Each role sees only what it may see

→

3 · AI jobs

- Tomorrow's BP estimate
- Ring night monitor
- Live early warning (every minute)
- Weekly doctor report

→

4 · Doctor

- Sees estimates, flags, weekly report
- Reviews flags
- Approves each patient's personal range

Rule we never break: the AI only informs the doctor. Alerts come from fixed medical rules in the backend; the AI never raises an alarm, diagnoses, or touches medication, and patients see no AI output.

AT A GLANCE

What exists today

24
database tables, row-level security on every one (52 policies)

6
AI models built, tested and documented (model cards)

4
real public datasets used for calibration and checks

95
automated tests, all passing

14
demo accounts in the cloud (12 patients, 1 doctor, 1 admin)

730
demo BP readings (simulated, labelled)

20,832
simulated ring samples: 12 patients × 28 nights

30
days of day-by-day testing on patients no model had seen

AI outputs already in the cloud database: 10 next-day BP estimates (+1 "not enough history"), 33 ring flags, 11 weekly doctor reports, 43 personal ranges proposed to the doctor.

DATABASE

One secure database for patients, doctors and AI

Area	What it holds
People	profiles, patient details (age, conditions, habits), doctor–patient assignments, body measurements
Blood pressure	readings with source (manual, CSV, device, simulated), validity check, duplicate protection
Daily life	lifestyle logs, medications and doses, symptoms, journal for the doctor
Ring	devices, ring samples (heart rate, HRV, SpO2, skin temperature)
Care	care plans (BP targets), alerts with episodes
AI outputs	BP estimates, ring flags and early warnings, weekly reports, personal ranges

Security and quality

- Row-level security on all 24 tables; nothing for logged-out users; public sign-up off
- Doctors see only their assigned patients
- Patients cannot read any AI output
- A trigger checks every reading (ranges, systolic > diastolic)
- Simulated data must be labelled "simulated"; the database rejects it otherwise

Built in 9 steps (migrations)

M1 schema · M2 access rules · M3 backend alignment · M4 AI predictions · M5 birth year · M6 Algerian risk factors + wearables · M7 AI monitoring · M8 chronic conditions · M9 live early warnings ALL IN CLOUD

DATA

Simulated where no real data exists, real where it does

Simulated (always labelled)

- **Demo patients:** 12 fictional patients with stories (sleep apnea, COPD, atrial fibrillation, rising BP)
- **Training cohort:** 200 patients × 90 days. Day-to-day BP variation matched to a **real** home-BP series (variation 10.0 vs 10.0 mmHg)
- **Unseen test cohort:** 100 new patients, used only for the 30-day test
- **Ring data:** built from published values (normal ranges, apnea, COPD)
- **Live nights:** 700 + 700 minute-by-minute nights with situations (oxygen falling, fever, bad sleep, loose ring…)

Real public data (statistics only, never uploaded)

Dataset	Used for
NHANES 2017–2020 (CDC, US adults)	population pulse / BP bands; risk model training
NHANES 2021–2023	external test on new real people
Mendeley home BP (1 patient, 128 readings)	made the simulator realistic
PhysioNet Apnea-ECG (8 real nights)	checked the ring oxygen rule

No open dataset has daily home BP for many patients. That is why the BP models are trained on calibrated simulation.

MODELS

Six models, each answering one question

Model	Question	Method	Status
mbp-v1.1	What will tomorrow morning's BP be?	ridge regression on the last 7 days + a 90 % range	SERVED doctor view
ring-v1.1	Was tonight unusual for this patient?	medical limits + the patient's own 28-day normal + IsolationForest	SERVED flags
wcls-v1.1	Will next week be above the BP target?	logistic regression	SERVED weekly report
live-v1	Will a limit be crossed in the next 20 minutes?	gradient boosting, checked on every new minute	BUILT not switched on yet
mbp-v2.1	Does ring data improve the BP estimate?	mbp + ring features	EXPERIMENT no real gain
wrisk-v1	Is BP uncontrolled despite treatment?	trained on 2,309 **real** people (NHANES)	HIDDEN too weak

All in Python (pandas, scikit-learn). Small, explainable models, each compared against a simple baseline. Patients are split so no test patient was seen in training.

NEXT-DAY BLOOD PRESSURE · MBP-V1.1

Beats the simple methods on every one of 30 test days

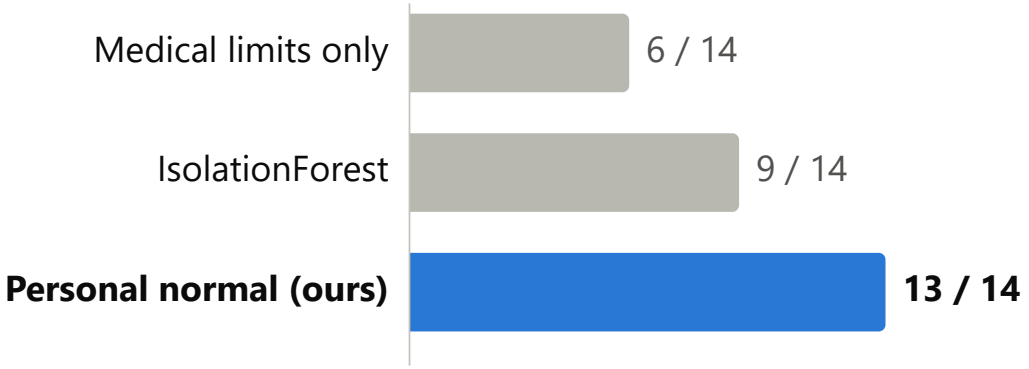


Systolic error, mean absolute (mmHg); lower is better. 2,271 patient-days, 2–31 March, 100 patients no model had seen.

- **Day-by-day test:** each day, the frozen model sees only earlier days, exactly as in real use
- Lower error than "same as yesterday" on **30 / 30** days, and than the 7-day mean on **28 / 30**
- Diastolic error 4.56 mmHg (7-day mean 4.83)
- The 90 % range contained the true value **90.7 %** of the time
- Shown to the doctor as "model estimate (synthetic training data)"

> Synthetic test patients. Real accuracy needs consented daily home readings from PULSE users.

The patient's own normal finds what fixed limits miss



Planted events found out of 14 (30-day test, 2,703 simulated nights). Other flags per patient-week: limits only 1.12, IsolationForest 0.08, personal normal 0.22.

- **Three checks each night:** medical limits (SpO2 < 94 %, resting heart rate > 100), the patient's own 28-day normal, and an anomaly detector
- **Conditions change the rules:** COPD → SpO2 limit 88 %; atrial fibrillation → HRV ignored
- **Real check (PhysioNet):** the oxygen rule flagged **4 / 4** real sleep-apnea nights and **0 / 3** healthy nights
- Flags go to the doctor to confirm or dismiss; they never create alerts

> Events were planted in simulated data. The real check is 8 nights from a sleep-lab oximeter, not a ring.

WEEKLY DOCTOR LOOP · WCLS-V1.1

Every week the doctor gets a report, and the doctor stays in charge

1 · Nights checked

ring flags written for the doctor

→

2 · Weekly report

home BP category, control vs target, chance next week is above target, pulse vs people of the same age and sex (NHANES)

→

3 · Model proposes

a personal range per ring metric

→

4 · Doctor decides

approves, edits or rejects; next nights are judged against the doctor's range

0.963

AUROC, wcls-v1.1 (30-day test, 400 predictions)

0.881

AUROC, "status stays the same"

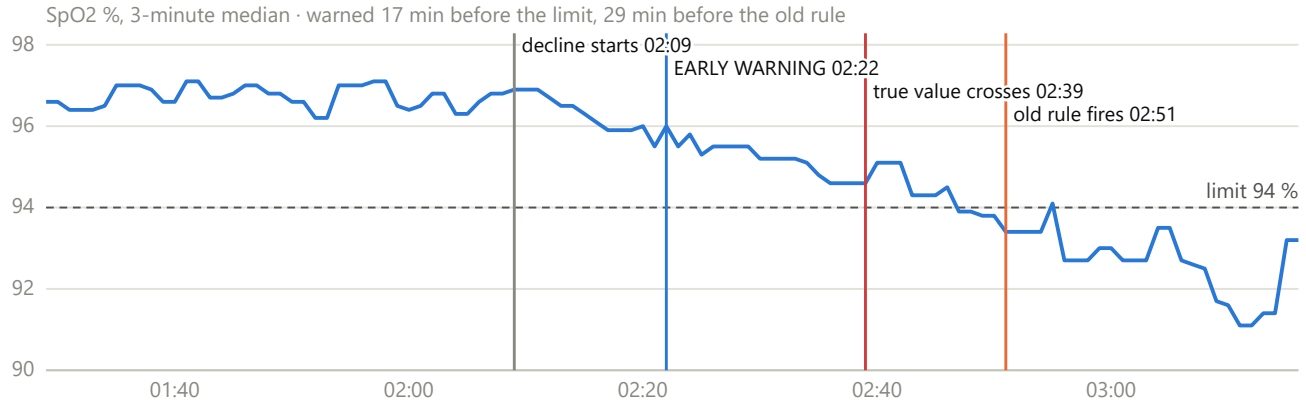
Honest limit: it ranks patients well, but it can't predict the weeks where the status *changes*. So it is shown as "how close to the target line", not as a forecast of change.

Doctor-reviewed flags become labels. A retuned model must still find the known events, and it is never deployed automatically.

LIVE EARLY WARNING · LIVE-V1

Warns the doctor 17–20 minutes *before* a limit is crossed

SIMULATOR-ONLY RESULTS: 700 + 700 simulated nights. Not real-world or clinical validation.



One unseen simulated night with falling oxygen, replayed minute by minute.

Situation	warned before	how early
oxygen falling	**93.5 %**	~17 min
heart rate climbing	**97.1 %**	~20 min
fever starting	**94.5 %**	~17 min

- The current "limit crossed" rule reacts 7–9 min **after**
- False warnings: about 1 in 25 normal nights
- Restless nights: 95 % warned, typically by 01:00
- About 44 ms per new minute per patient

HONESTY ABOUT REAL DATA

When a model was too weak on real people, we hid it

wrisk-v1 on real NHANES data

- Trained on **2,309 real** treated US adults: can risk factors tell whose BP is uncontrolled?
- Tested on a later survey it never saw (2021–2023, **1,851 people**)
- AUROC **0.596** vs 0.565 for age + sex alone: stable, but too weak
- Not shown to doctors, because a near-random score would mislead

> **Lesson:** risk factors can't tell whose BP is controlled. Control has to be **measured**, which is exactly what PULSE does.

What is and isn't validated

Claim	Evidence
Simulator realism	matched to a real home-BP series
Ring oxygen rule	8 real sleep-lab nights
Next-day BP, weekly classifier	unseen **simulated** patients only
Live early warning	**simulated** nights only

Next step for every model: consented real PULSE data, re-test by patient and over time, then drift monitoring.

SAFETY BY DESIGN

Designed so the AI never acts on its own

- **Advisory only:** AI output goes to the doctor, never to the patient (enforced in the database)
- **Fixed rules decide alerts** in the backend: BP ≥ 180/120 and red-flag symptoms; ring rule R-RING-1 proposed
- **Emergency number 14** (Protection civile); never dialled in tests or the demo
- **No medication advice** anywhere
- **Every simulated row is labelled**; real datasets used as statistics only, under their licences
- Every model has a model card: data, method, results, limits

Clinician review

DONE Medical mentor, 30 Oct: alert thresholds 180/120, red flags, SpO2 94 % (COPD 88 %), band rule, patient wording

DONE Medical mentor, 30 Oct: live-v1 thresholds and timings (P thresholds HR 0.35 / SpO2 0.45 / temperature 0.85, 20-min horizon, 2 minutes in a row, 30-min quiet period, 2-min freshness, restless P ≥ 0.7), ring personal limit z 3.5, R-RING-1

Reviewed for prototype use. The review covers the settings; the live results stay simulator-only.

STATUS

What is done, what is waiting, what is missing

Status	Item	Who
LIVE	Cloud database (M1–M9), demo data, AI outputs (estimates, flags, reports, proposed ranges)	Data/AI
BUILT	6 models, daily / nightly / weekly jobs, live engine, 95 tests, model cards, full 21-page report	Data/AI
DONE	Clinician review of every threshold; demo password and backend message sent	Mentor, you
ON HOLD	Cloud AI jobs, live engine, simulators and R-RING-1 stay off until Ayoub lifts the freeze	Ayoub
TO BUILD	Backend: alert rules writing to Supabase; endpoints for weekly report, flag review (incl. early warnings), range approval; R-RING-1; set number 14 as verified	Backend
TO BUILD	Frontend: doctor panels for estimate, ring flags, early warnings, weekly report, range approval	Frontend
MISSING	A real device (validated upper-arm cuff, ring) and its adapter; a server to run the live monitor 24/7	Team
MISSING	Real, consented patient data, needed to measure real accuracy	Team, later

FOR YOU PRESENTER ONLY · SKIP WHEN PROJECTING

What you need to do

Done

The mentor accepted every threshold · demo password shared privately · backend message sent · M9 is in the cloud and Ayoub decided to keep it (31 Oct)

Still to do

1. **Respect Ayoub's freeze:** no cloud scripts, AI jobs or simulators on the cloud until he lifts it. Ask him what he needs before live-v1 can be switched on.
2. **Review the pull request** `fix/data-m9-record` with Ayoub: it aligns the data docs with his written decision.
3. **Team decision:** which real device (validated upper-arm cuff recommended).
4. If the review note had items 1–3, paste them to Claude.

Optional live demo (your laptop)

1. Start **Docker Desktop** and wait until it's running (the local database starts with it).
2. Terminal 1, in `ml\`:
   `..\.venv\Scripts\python -m pulse_ml.monitor --follow --every 5`
3. Terminal 2, in the repo root:
   `.venv\Scripts\python -m pulse_data.stream --patient-email patient.b@demo.pulse.test --scenario oxygen_falling --speed 5 --minutes 20`
4. An `early_warning` row appears before the "limit crossed" flag.

Both terminals: `DATABASE_URL` = the local dev stack (runbook step 2.3). Simulated rows, **local** only; the cloud is refused.

Files to open if asked

`docs/data/PULSE_FULL_REPORT.pdf` (every detail) · `ml/MODEL_CARD.md`

FOR YOU PRESENTER ONLY

Likely questions, short answers

Question	Answer
Is this real patient data?	No. All patient data is simulated and labelled. Real public data was used to make the simulator realistic and to check the rules.
How accurate is it?	On 100 unseen simulated patients, next-day systolic error is 7.9 mmHg, better than simple methods on every one of 30 days. Real accuracy needs consented PULSE data.
Can the AI raise an alarm?	No. Alerts come from fixed medical rules in the backend. The AI only informs the doctor; patients never see it.
Why not deep learning?	Little data, and doctors need to understand the output. Small models beat simple baselines and are easy to explain.
You have no ring. How do you know it works?	Ring data is simulated from published values, and the oxygen rule was checked on 8 real sleep-lab nights (4/4 apnea flagged, 0/3 healthy).
Did any model fail?	Yes: the real-data risk model (AUROC 0.60) was too weak, so we hid it. A first live forecaster never warned; we replaced it.
What happens next?	Real device, consented data, re-test every model on real people, then drift monitoring. The thresholds are already clinician-reviewed for the prototype.